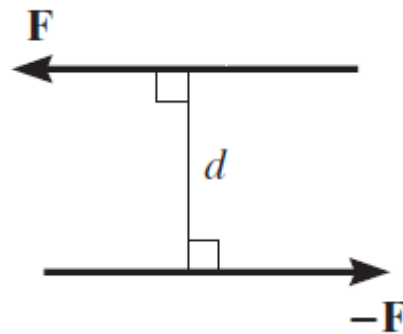


Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)
Lecture 6: Force System
Resultants (Part 2)
By Saeed Lotfan

Moment of a Couple

Couple: A *couple* is defined as two parallel forces that have the same magnitude, but opposite directions, and are separated by a perpendicular distance d .

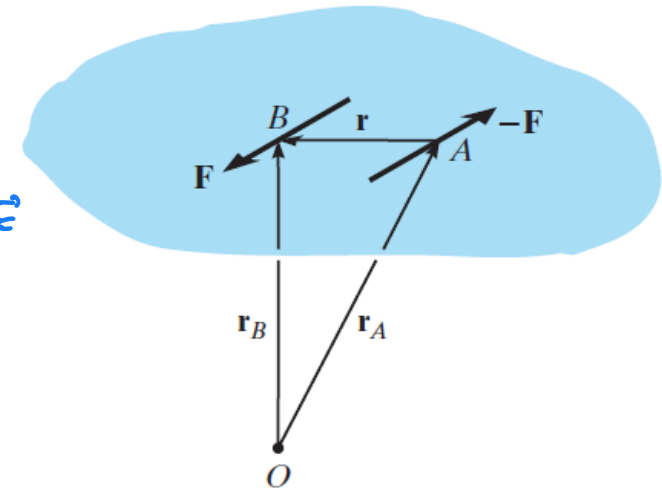


Since the resultant force is zero, the only effect of a couple is to produce a rotation, or if no movement is possible, there is a tendency for rotation.

Moment of a Couple

The moment produced by a couple is called a ***couple moment***.

$$\vec{M}_O = \vec{r} \times \vec{F}$$

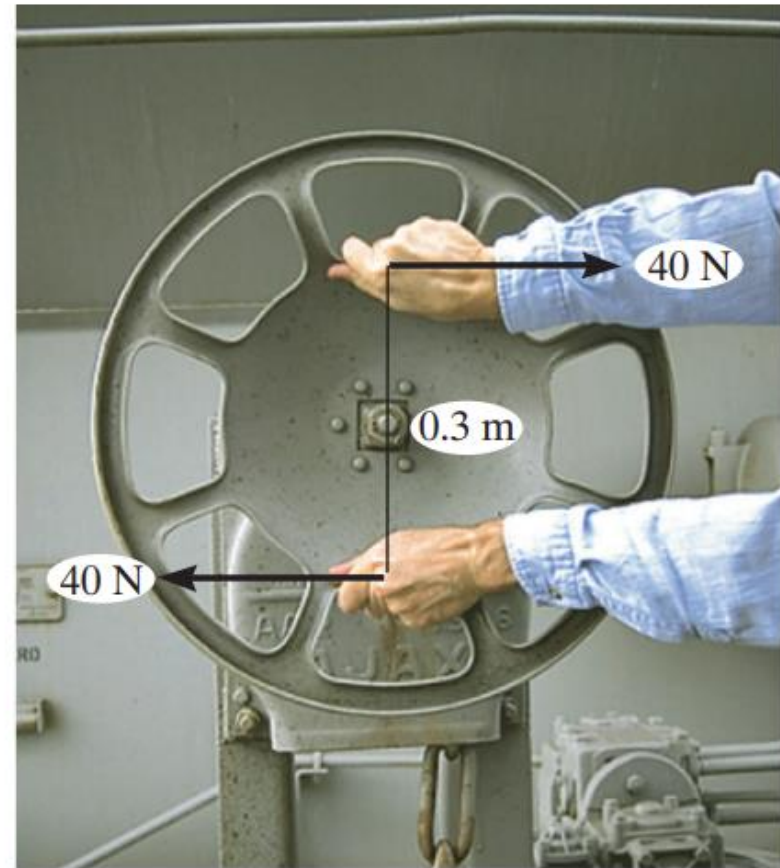
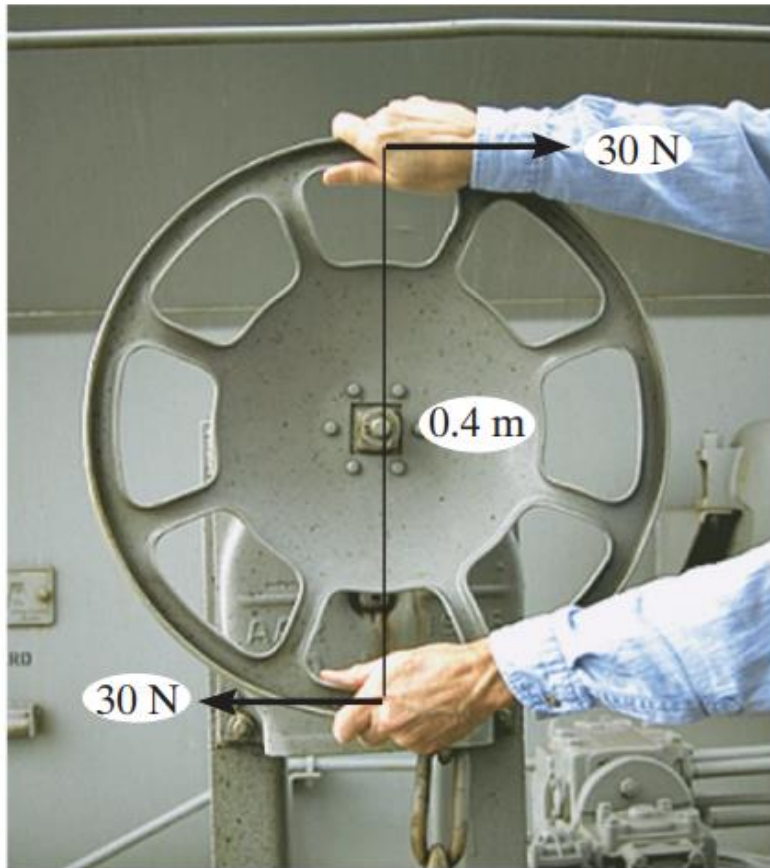


$$\mathbf{M} = \mathbf{r}_B \times \mathbf{F} + \mathbf{r}_A \times (-\mathbf{F}) = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{F}$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}$$

This result indicates that a couple moment is a ***free vector***, since it can act at any point since $\mathbf{M}$ depends only upon the position vector $\mathbf{r}$ directed between the forces and not the position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$, directed from point O to the forces.

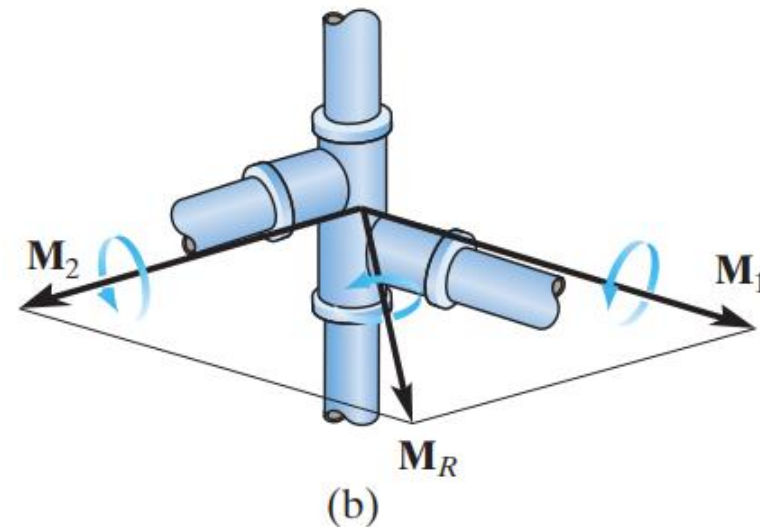
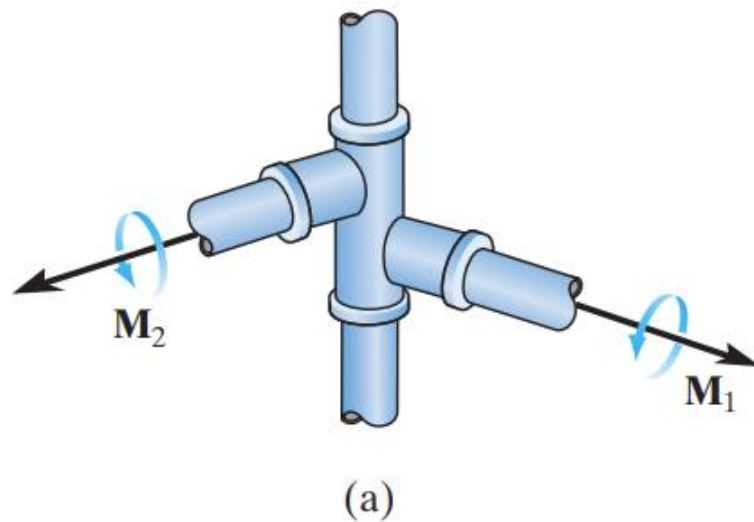
Moment of a Couple



If two couples produce a moment with the same magnitude and direction, then these two couples are equivalent.

Resultant Couple Moment

Since couple moments are free vectors, their resultant can be determined by moving them to a single point and using vector addition.



Concluding Notes

- A couple moment is produced by two noncollinear forces that are equal in magnitude but opposite in direction. Its effect is to produce pure rotation, or tendency for rotation in a specified direction.
- A couple moment is a free vector, and as a result it causes the same rotational effect on a body regardless of where the couple moment is applied to the body.
- The moment of the two couple forces can be determined about *any point*. For convenience, this point is often chosen on the line of action of one of the forces in order to eliminate the moment of this force about the point.
- In three dimensions the couple moment is often determined using the vector formulation, $\mathbf{M} = \mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is directed from *any point* on the line of action of one of the forces to *any point* on the line of action of the other force $\mathbf{F}$.
- A resultant couple moment is simply the vector sum of all the couple moments of the system.

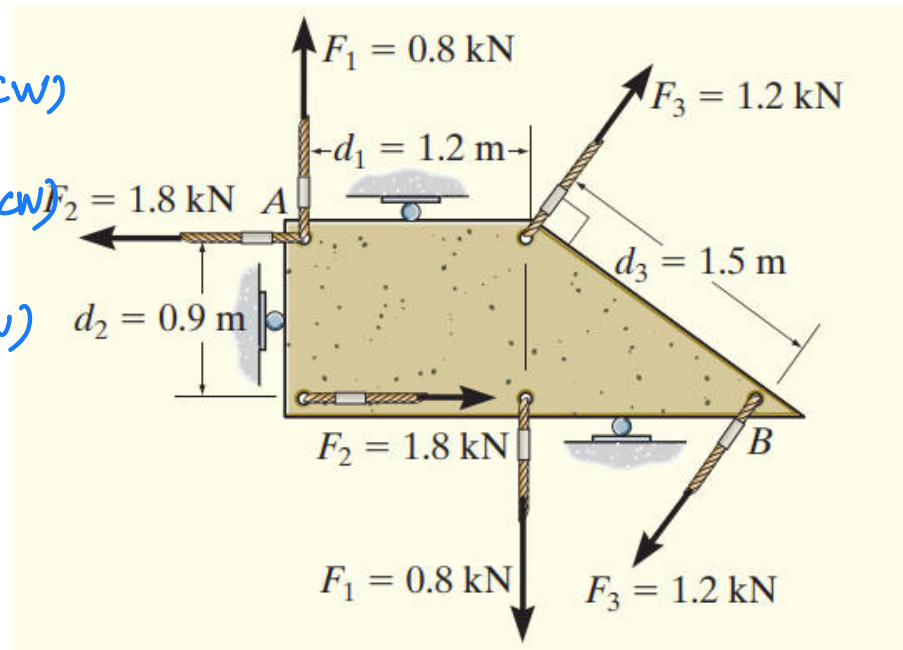
Example: Determine the resultant couple moment of the three couples acting on the plate shown below.

$$M_1 = 0.8 \times 1.2 \text{ kN}\cdot\text{m (CW)}$$

$$M_2 = 1.8 \times 0.9 \text{ kN}\cdot\text{m (CCW)}$$

$$M_3 = 1.2 \times 1.5 \text{ kN}\cdot\text{m (CW)}$$

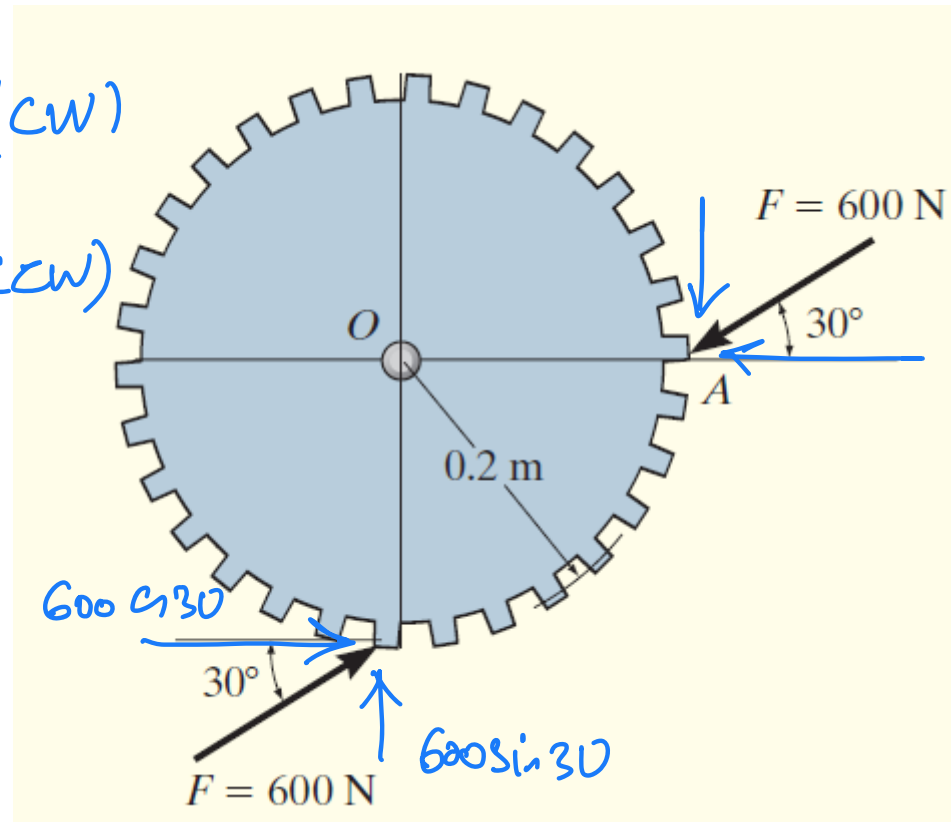
$$M_R = -M_1 + M_2 - M_3$$



Example: Determine the magnitude and direction of the couple moment acting on the gear shown below.

$$(600 \sin 30^\circ) 0.2 \text{ (CW)}$$

$$(600 \cos 30^\circ) 0.2 \text{ (CCW)}$$



Simplification of a Force and Couple System

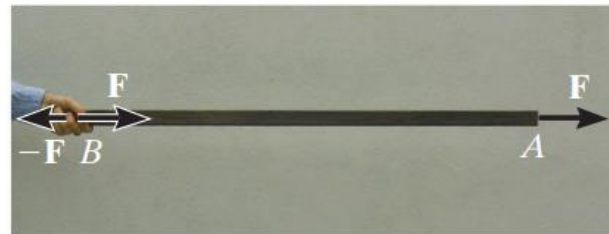
Sometimes it is convenient to reduce a system of forces and couple moments acting on a body to a simpler form by replacing it with an **equivalent system**, consisting of a single resultant force and a resultant couple moment. A system is equivalent if the **external effects** it produces on a body are the same as those caused by the original force and couple moment system. If the body is free to move, then the external effects of a system refer to the **translating and rotating motion** of the body, or if the body is fully supported, they refer to the **reactive forces** at the supports.

Simplification of a Force and Couple System



(a)

||



(b)

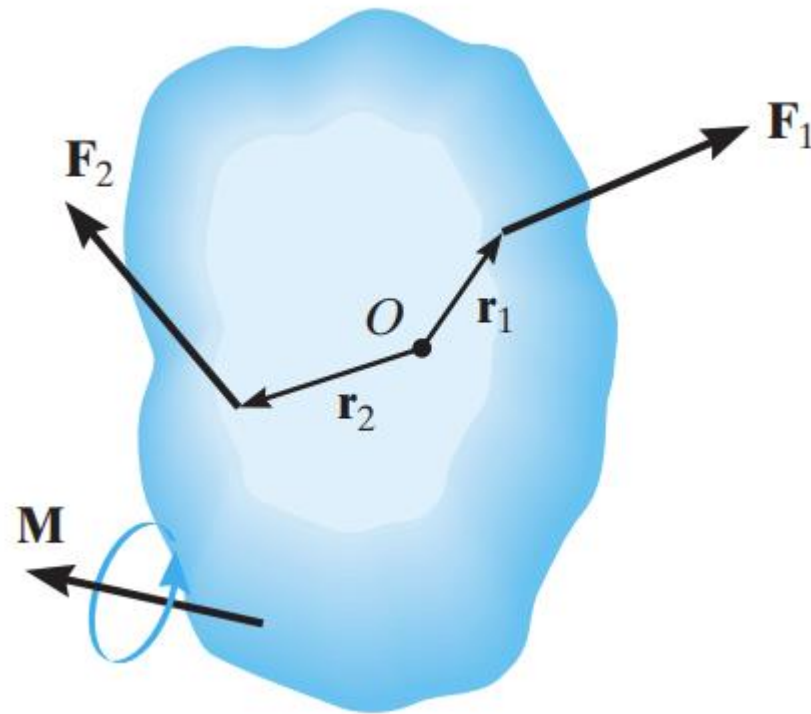
||



(c)

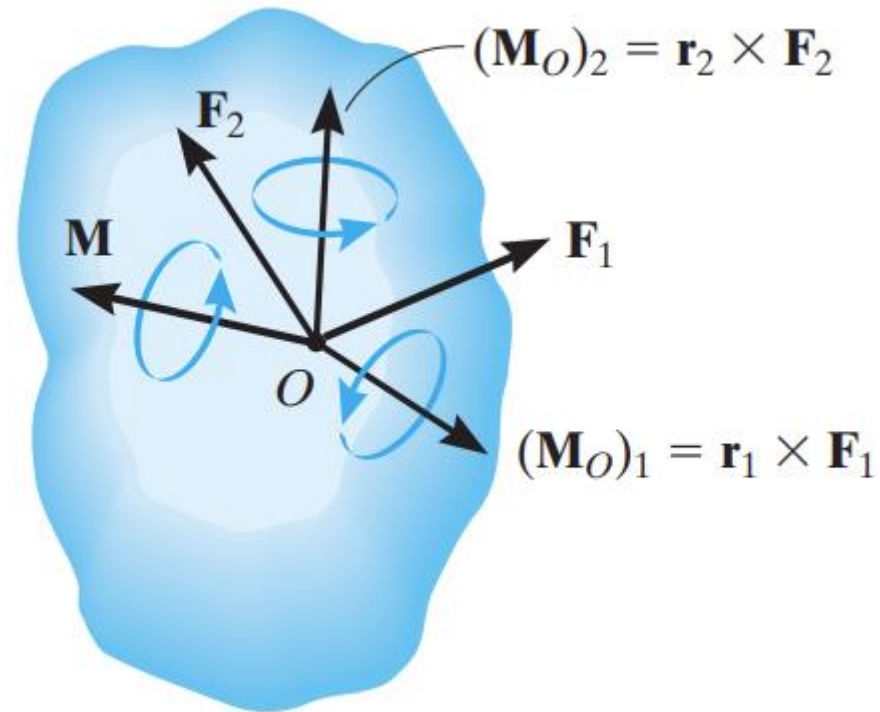
Simplification of a Force and Couple System

System of Forces and Couple Moments



Simplification of a Force and Couple System

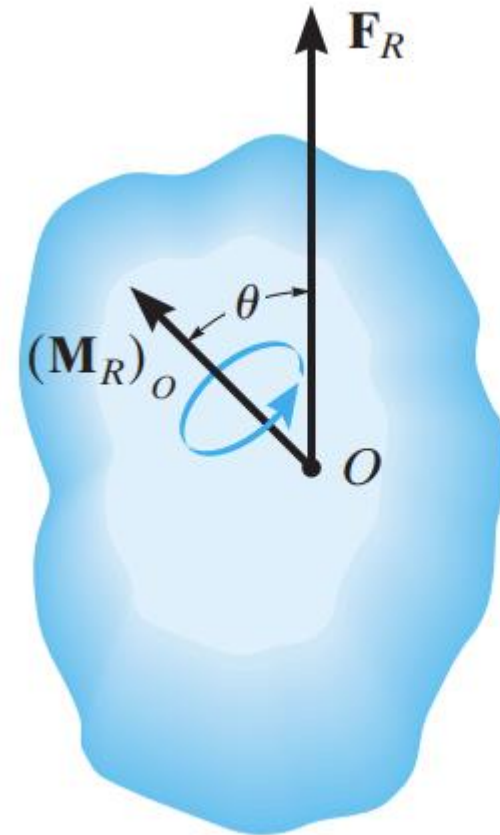
System of Forces and Couple Moments



Simplification of a Force and Couple System

System of Forces and Couple Moments

$$\mathbf{F}_R = \Sigma \mathbf{F}$$
$$(\mathbf{M}_R)_O = \Sigma \mathbf{M}_O + \Sigma \mathbf{M}$$



Simplification of a Force and Couple System

System of Forces and Couple Moments

The weights of these traffic lights can be replaced by their equivalent resultant force:

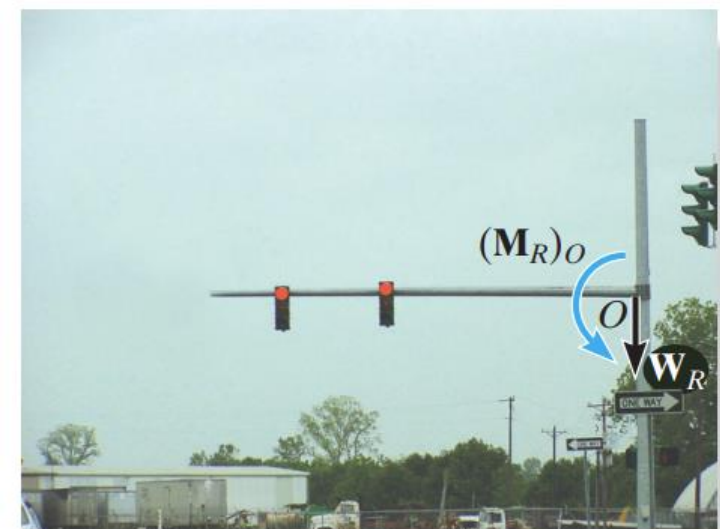
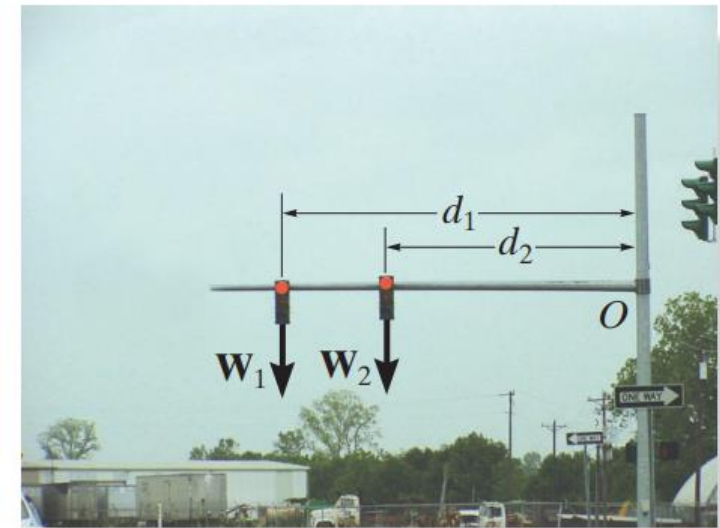
$$W_R = W_1 + W_2$$

and a couple moment:

$$(M_R)_O = W(d_1) + W_2(d_2)$$

at the support, O.

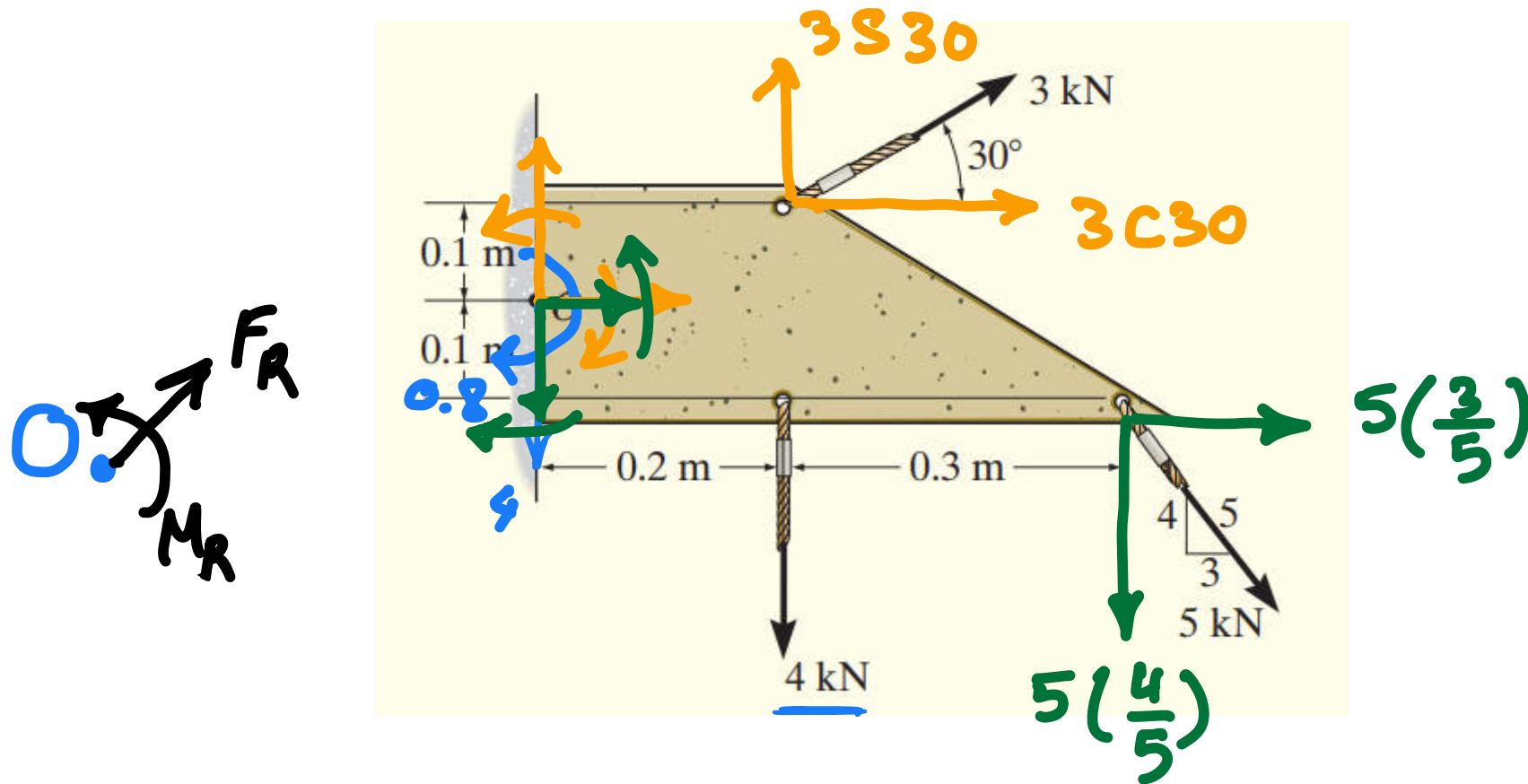
In both cases the support must provide the same resistance to translation and rotation in order to keep the member in the horizontal position.



Concluding Notes

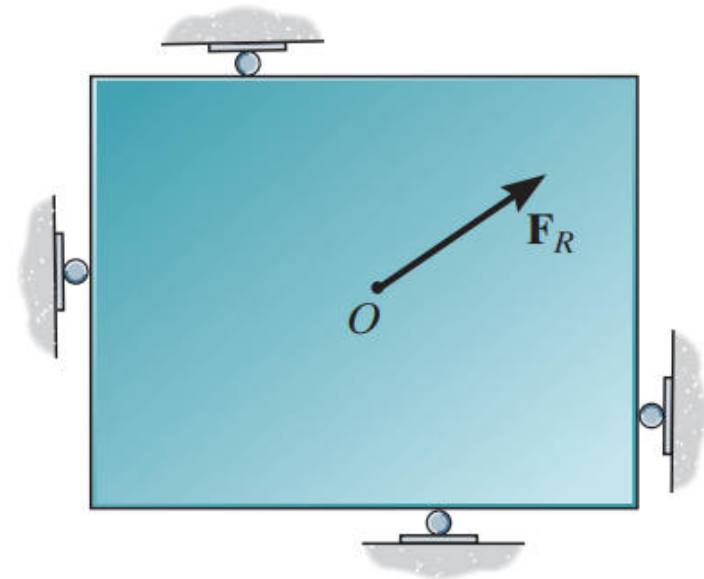
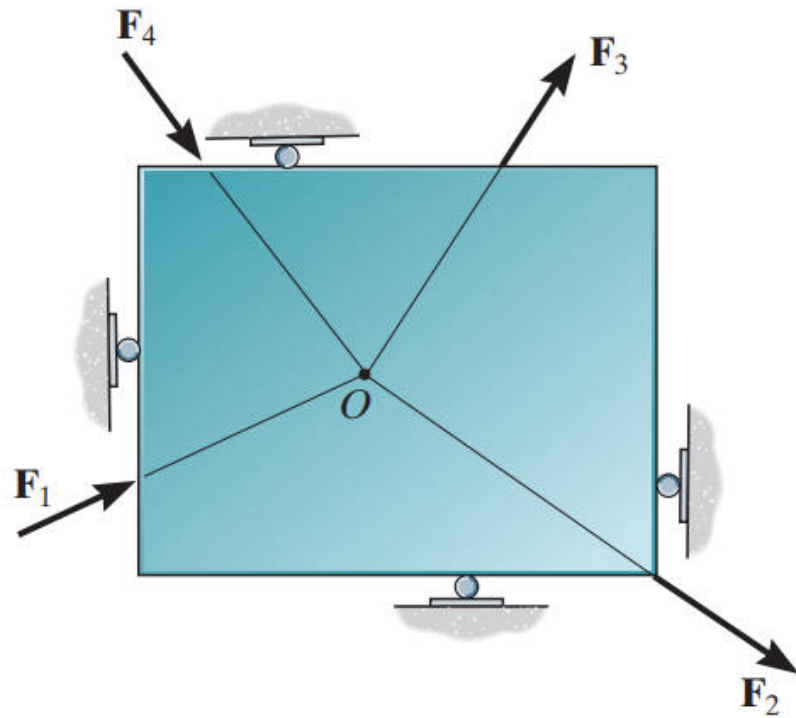
- Force is a sliding vector, since it will create the same external effects on a body when it is applied at any point P along its line of action. This is called the principle of transmissibility.
- A couple moment is a free vector since it will create the same external effects on a body when it is applied at any point P on the body.
- When a force is moved to another point P that is not on its line of action, it will create the same external effects on the body if a couple moment is also applied to the body. The couple moment is determined by taking the moment of the force about point P .

Example: Replace the force and couple system shown below by an equivalent resultant force and couple moment acting at point O.



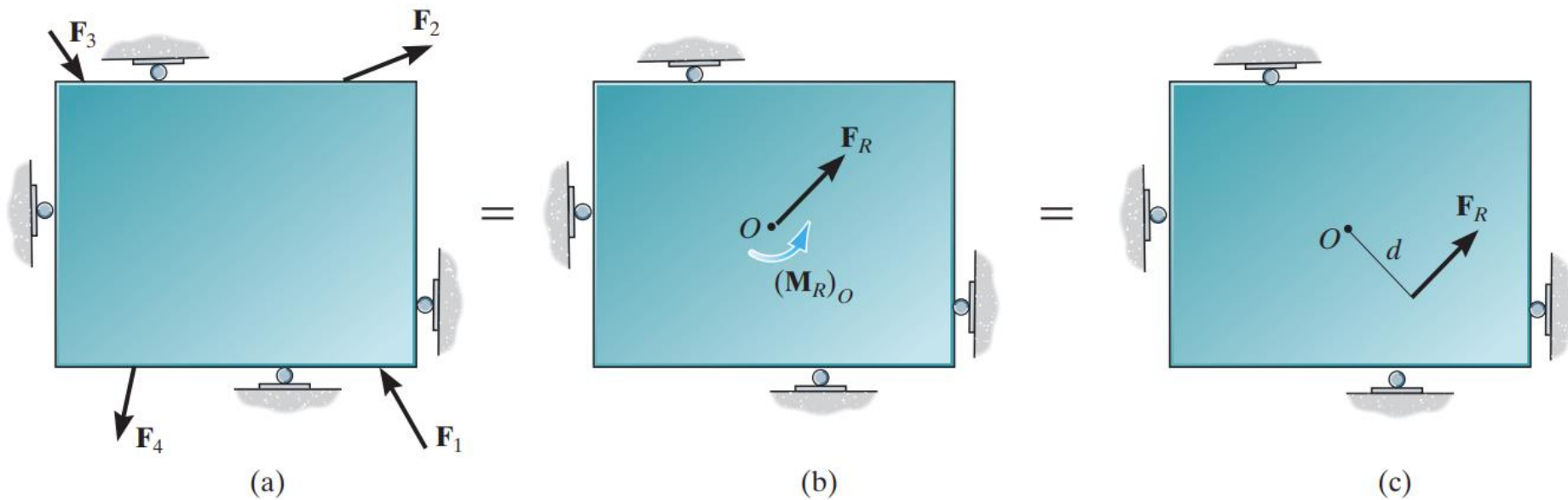
Further Simplification of a Force and Couple System

Concurrent Force System



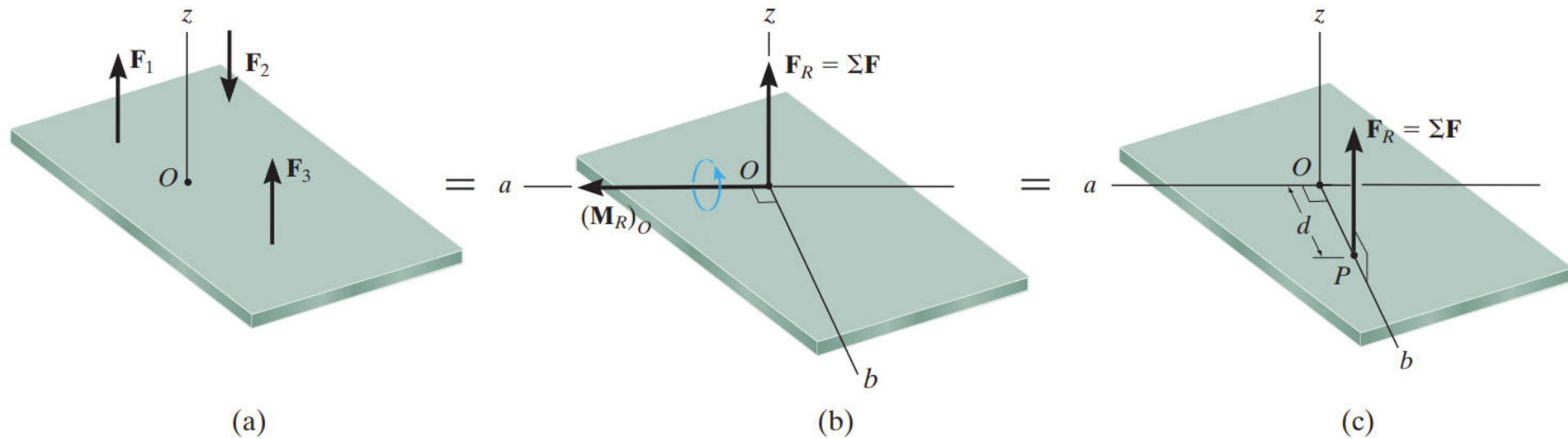
Further Simplification of a Force and Couple System

Coplanar Force System



Further Simplification of a Force and Couple System

Parallel Force System



Further Simplification of a Force and Couple System

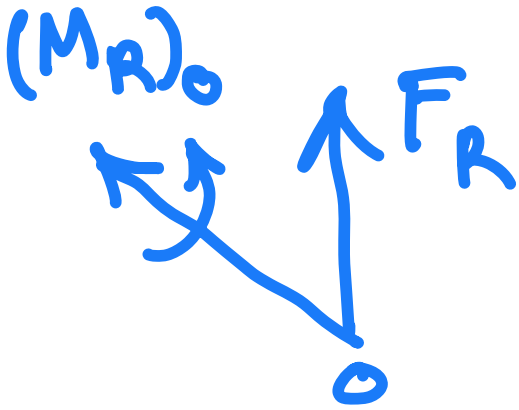
Parallel Force System



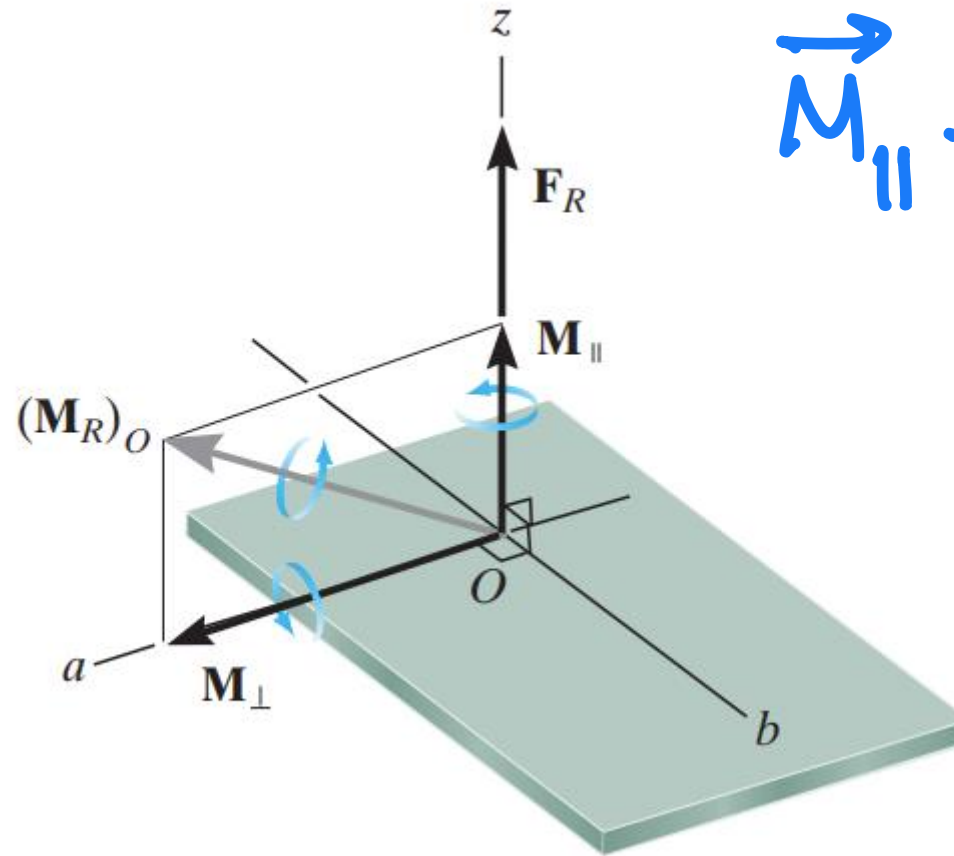
Here the weights of the traffic lights are replaced by their resultant force $W_R = W_1 + W_2$, which acts at a distance $d = (W_1 d_1 + W_2 d_2) / W_R$ from O . Both systems are equivalent.

Further Simplification of a Force and Couple System

Reduction to a Wrench (Screw)

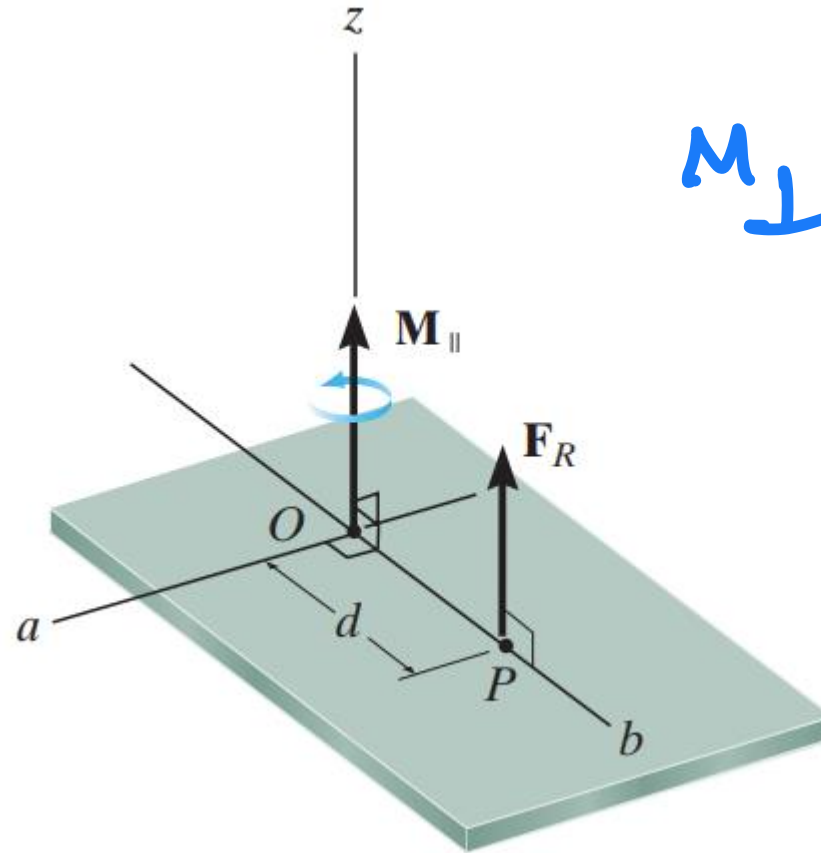


$$\vec{M}_{||} + \vec{M}_{\perp} = (\vec{M}_R)_O$$



Further Simplification of a Force and Couple System

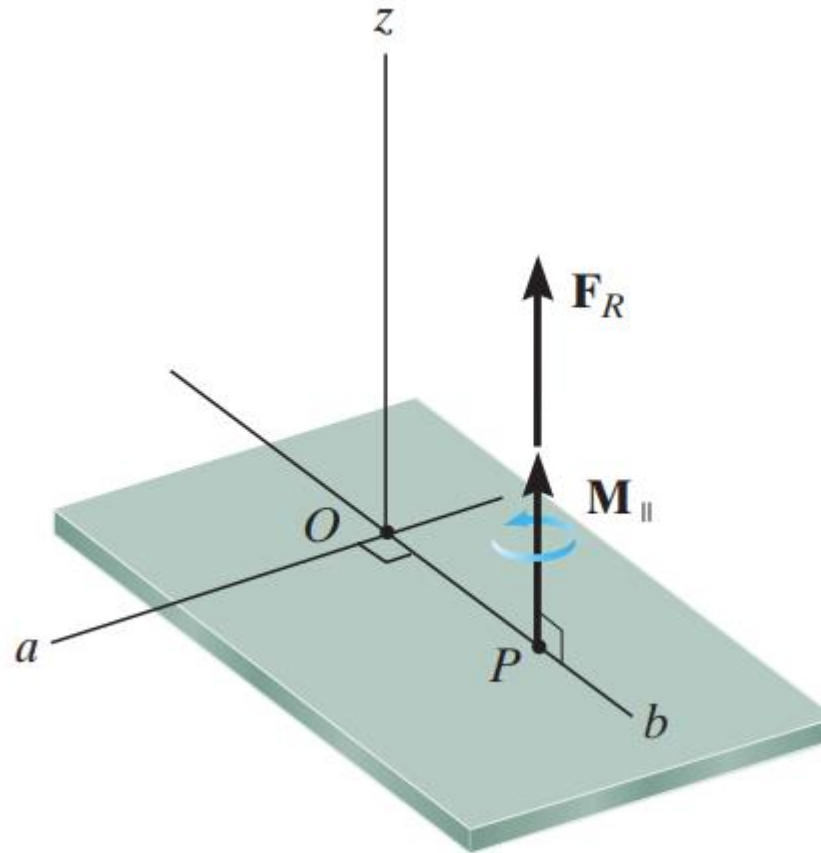
Reduction to a Wrench (Screw)



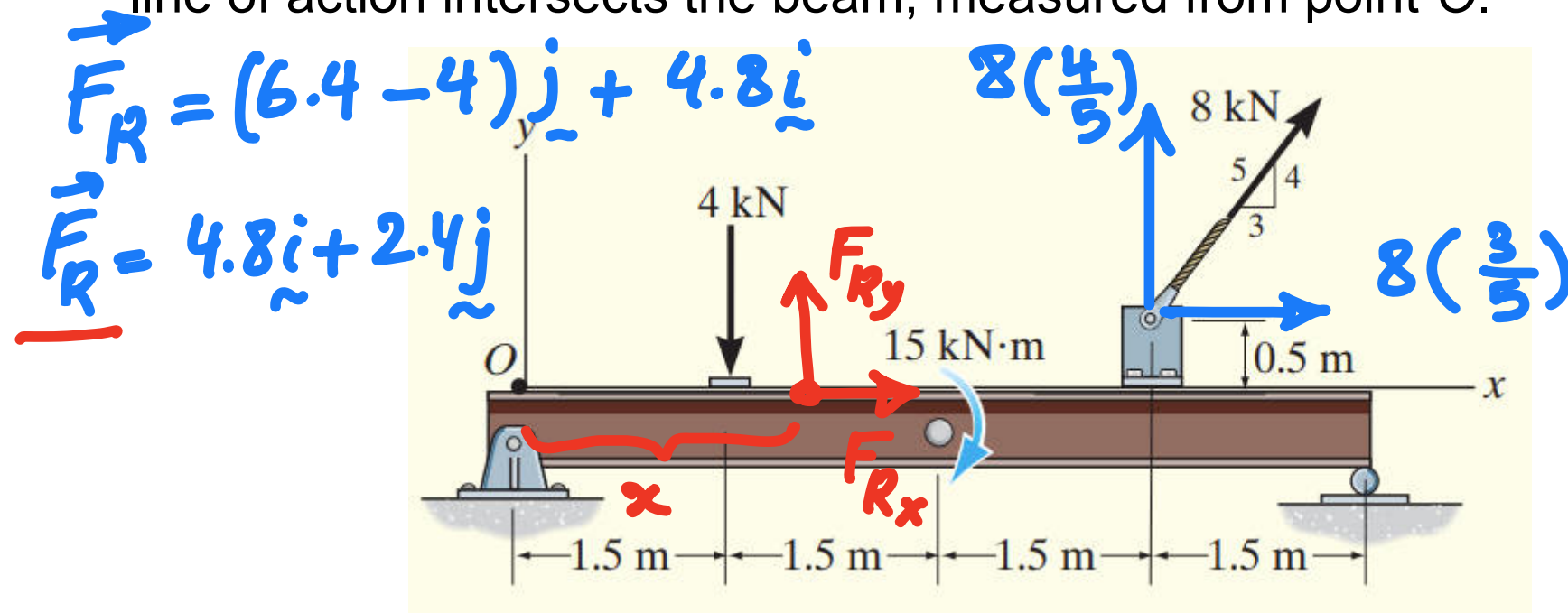
$$M_{\perp} = F_R d$$

Further Simplification of a Force and Couple System

Reduction to a Wrench (Screw)

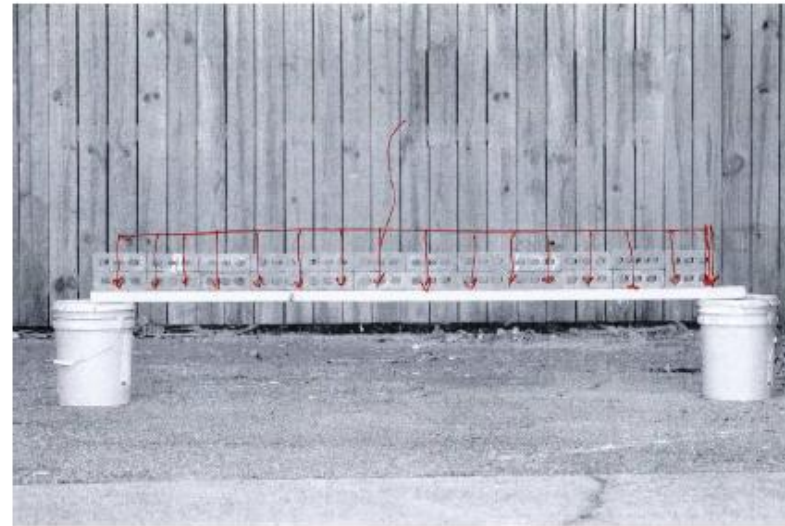
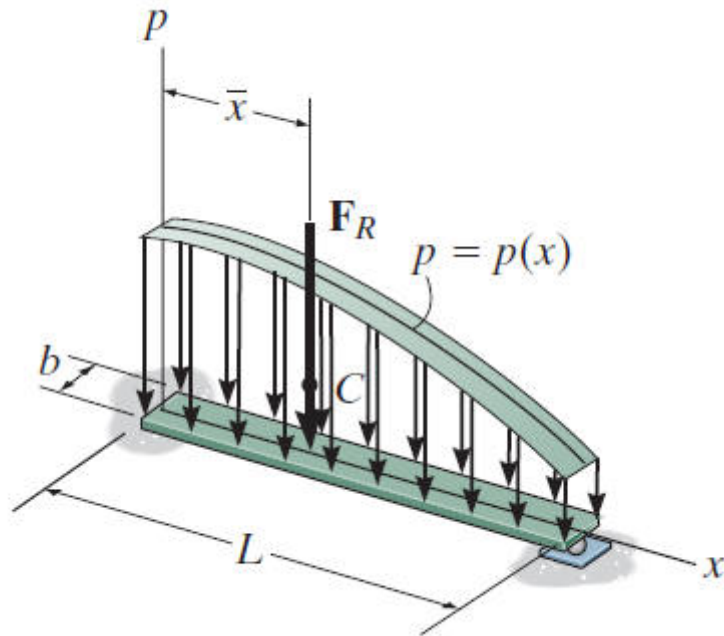


Example: Replace the force and couple moment system acting on the beam in figure below by an equivalent resultant force, and find where its line of action intersects the beam, measured from point O.



$$\begin{aligned}
 \sum M_O &\Rightarrow -4(1.5) + 8(\frac{4}{5})(4.5) - 8(\frac{3}{5})(0.5) - 15 \\
 &= F_{Ry} x
 \end{aligned}$$

Reduction of a Simple Distributed Loading



The pile of bricks creates a uniform distributed loading on the board.

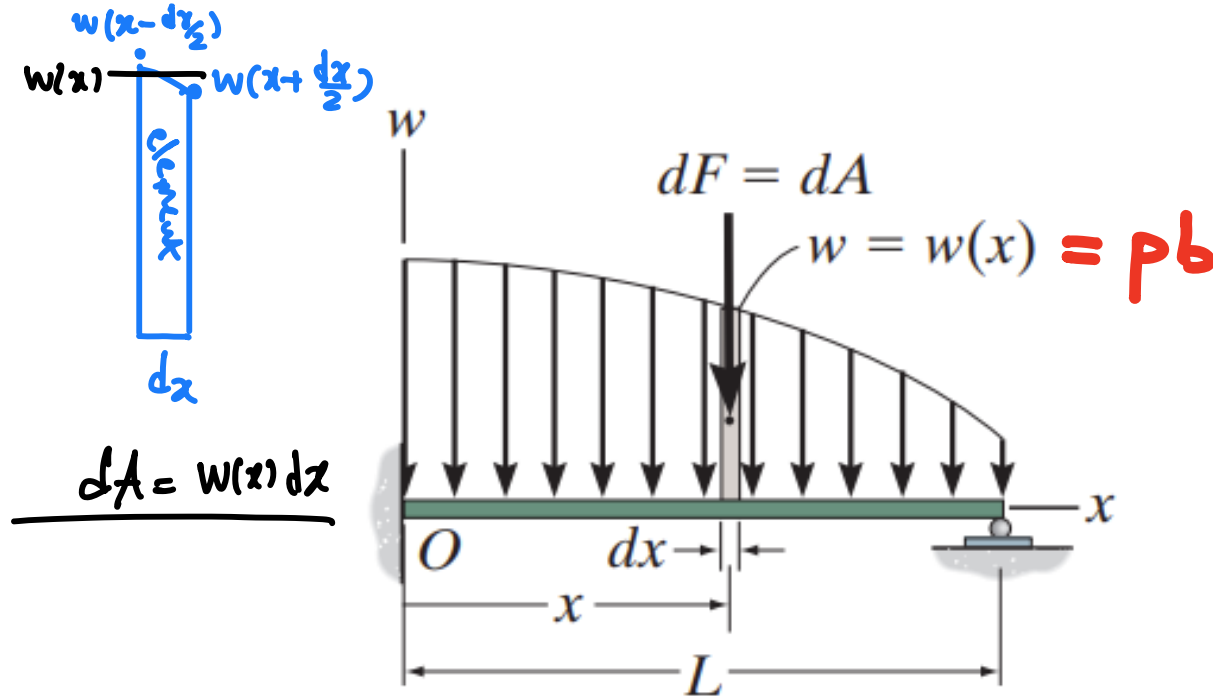
For example, wind on the face of a sign, water within a tank, or the weight of sand on the floor of a storage container all exert **distributed loadings**.

Reduction of a Simple Distributed Loading

Loading Along a Single Axis

$$\Delta F = W(x) \Delta x \rightarrow F_R = \sum_{i=1}^N W(x_i) \Delta x_i$$

$$F_R = \int_0^L W(x) dx$$



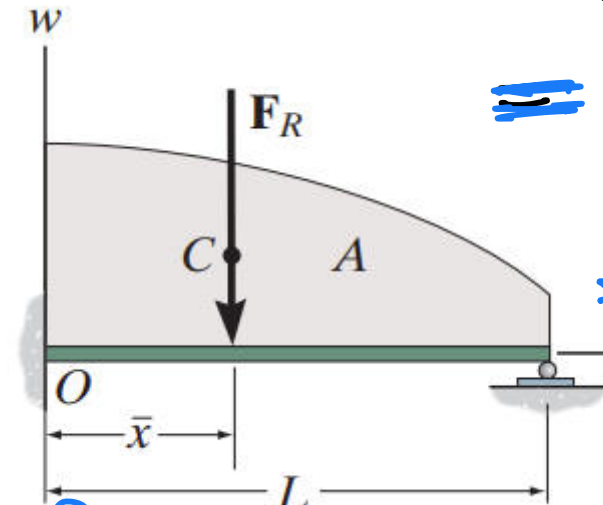
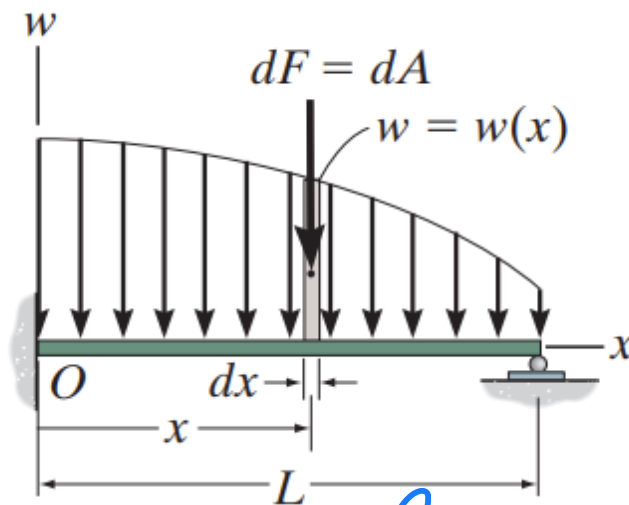
$$dA = w(x) dx$$

$$w(x) = p(x)b \text{ N/m}$$

Reduction of a Simple Distributed Loading

Magnitude

$$dM_o = x dF = x w(x) dx \rightarrow M_o = \int_0^L w(x) x dx$$



$$= F_R \bar{x}$$

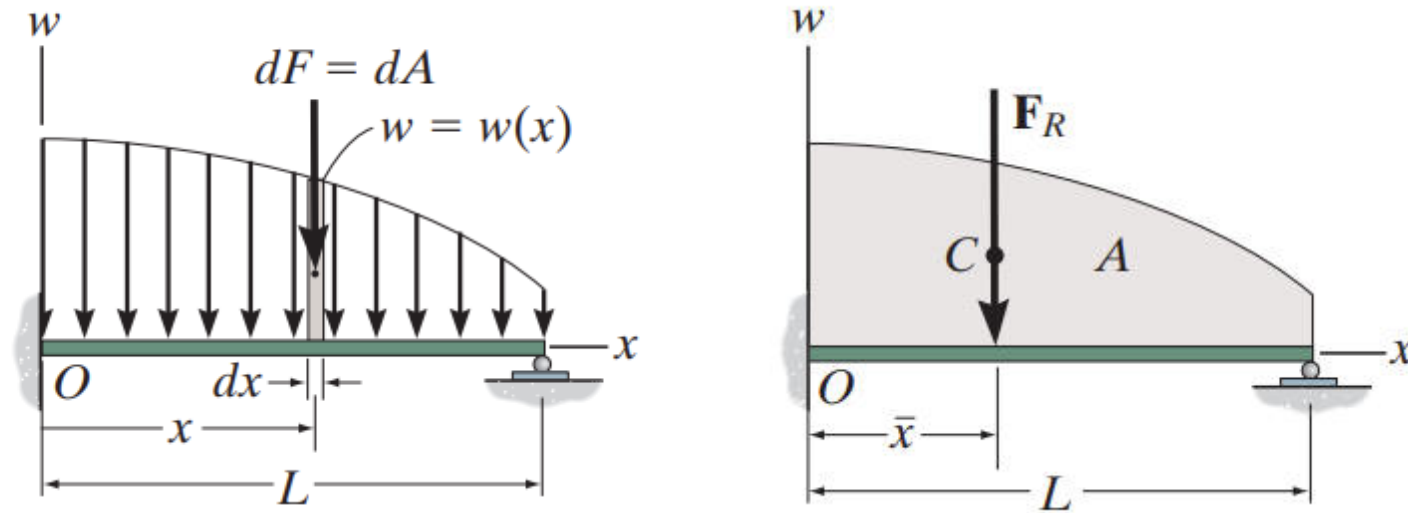
$$= \left(\int w(x) dx \right) \bar{x}$$

$$\bar{x} \int w(x) dx = \int x w(x) dx$$

$$F_R = \int_L w(x) dx = \int_A dA = A$$

Reduction of a Simple Distributed Loading

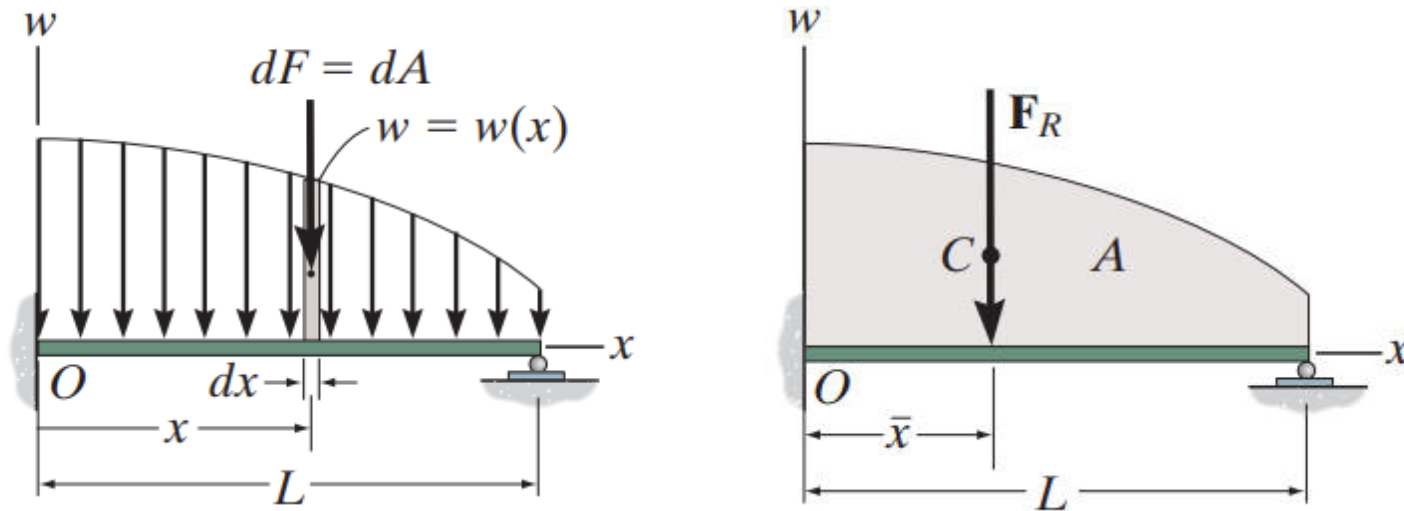
Location



$$\bar{x} = \frac{\int_L x w(x) dx}{\int_L w(x) dx} = \frac{\int_A x dA}{\int_A dA}$$

Reduction of a Simple Distributed Loading

Location

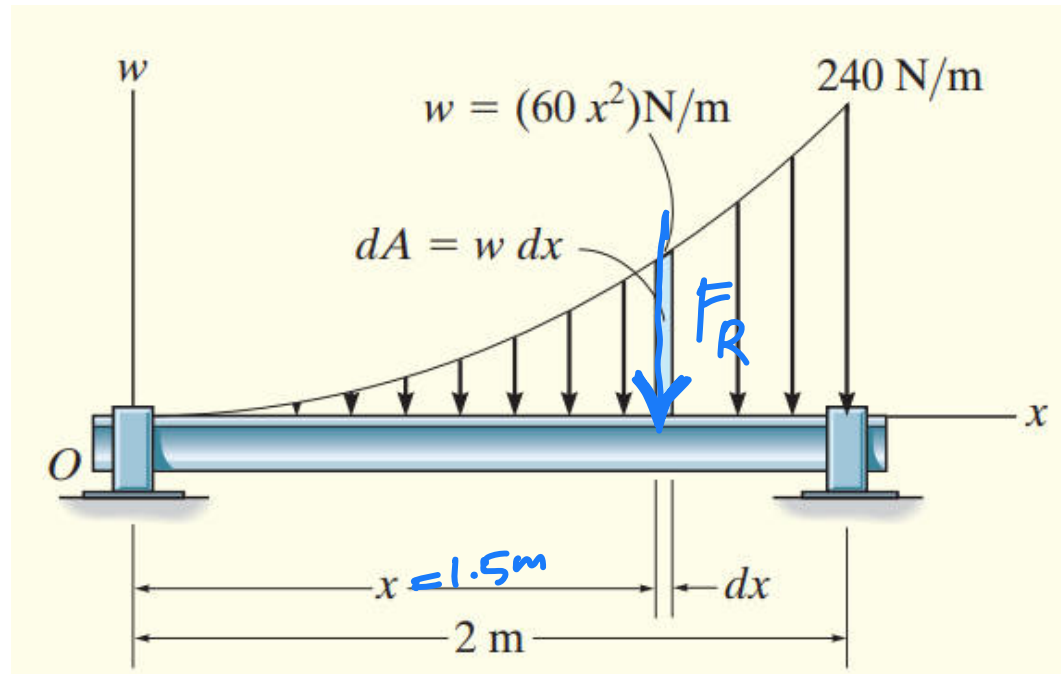


This coordinate $\bar{x}$ locates the **geometric center** or **centroid** of the area under the distributed loading. In other words, the line of action of the resultant force passes through the centroid C (geometric center) of the area under the loading diagram.

Concluding Notes

- Coplanar distributed loadings are defined by using a loading function $w(x)$ that indicates the intensity of the loading along the length of a member. This intensity is measured in N/m.
- The external effects caused by a coplanar distributed load acting on a member can be represented by a resultant force.
- This resultant is equivalent to the *area* under the loading diagram and has a line of action that passes through the *centroid* or geometric center of this area.

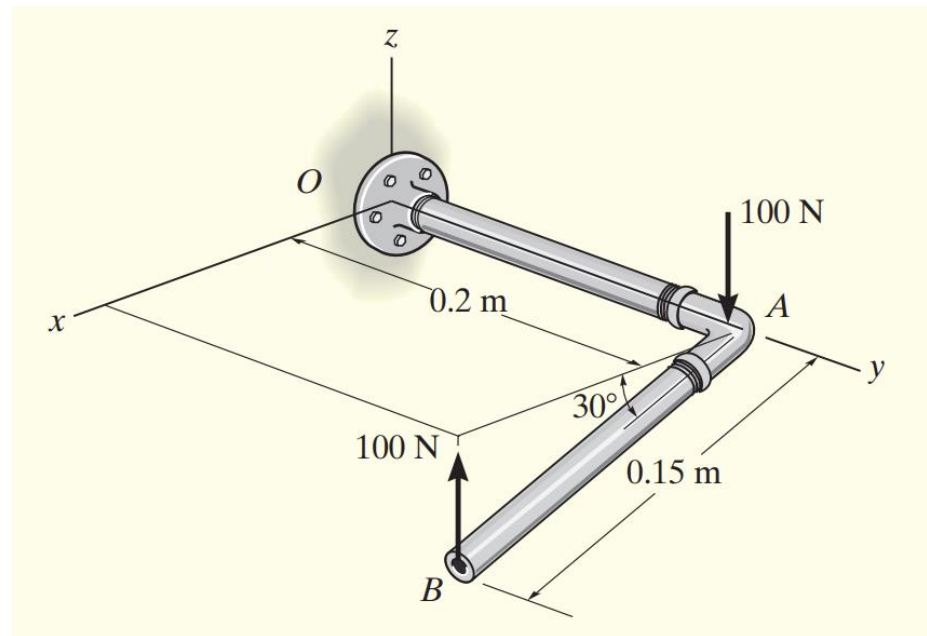
Example: Determine the magnitude and location of the equivalent resultant force acting on the shaft below.



$$\begin{aligned}
 dA &= w(x) dx \\
 F_R &= \int_0^L w(x) dx = \int_0^2 60x^2 dx \\
 &= 20x^3 \Big|_0^2 = 160 \text{ N} \\
 \bar{x} &= \frac{\int x w(x) dx}{F_R} \\
 &= \frac{\int 60x^3 dx}{160} = \frac{15x^4}{160} \Big|_0^2 \\
 &= \frac{15 \times 16}{160} = 1.5 \text{ m}
 \end{aligned}$$

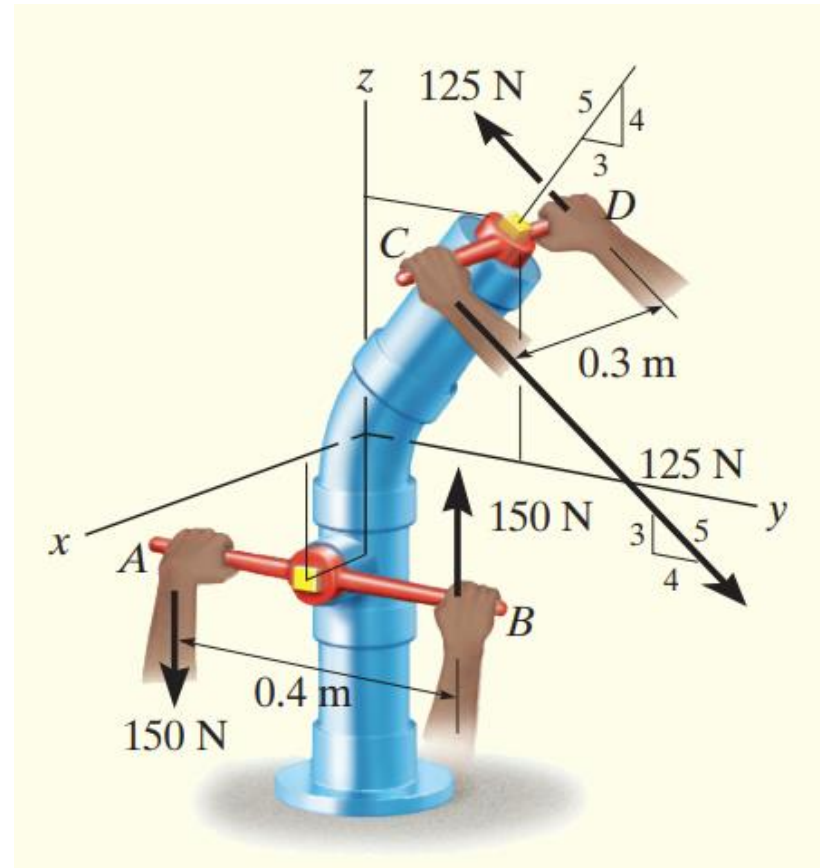
Practice at Home

1. Determine the couple moment acting on the pipe shown below. Segment AB is directed 30° below the x - y plane.



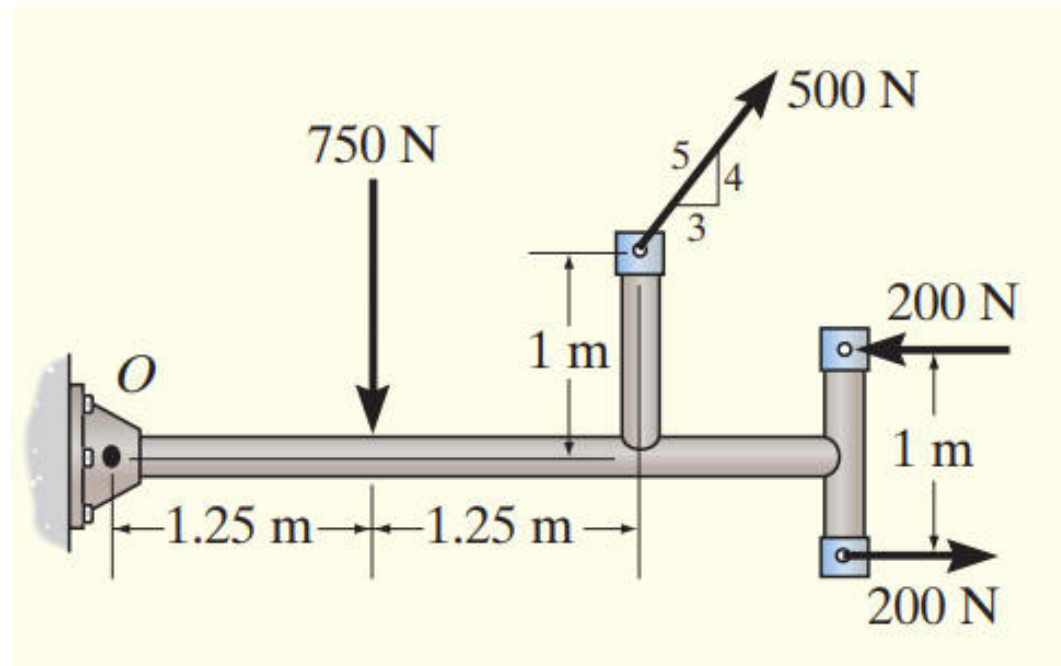
Practice at Home

2. Replace the two couples acting on the pipe assembly shown below by a resultant couple moment.



Practice at Home

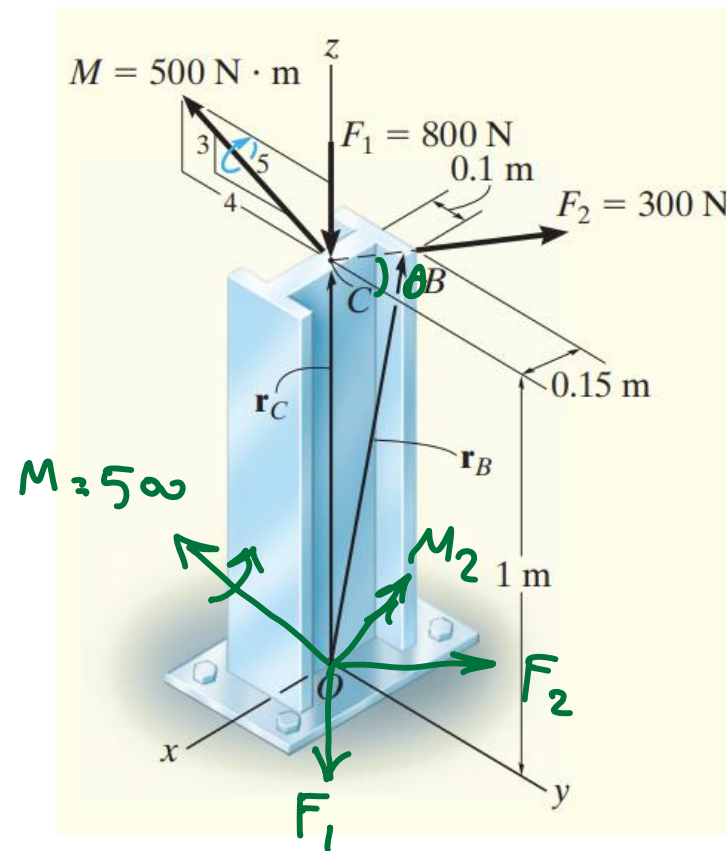
3. Replace the force and couple system acting on the member in the following figure by an equivalent resultant force and couple moment acting at point O .



Practice at Home

4. The structural member is subjected to a couple moment $\mathbf{M}$ and forces $\mathbf{F}_1$ and $\mathbf{F}_2$ below. Replace this system by an equivalent resultant force and couple moment acting at its base, point O.

$$\begin{aligned}\vec{M}_{F_2} &= \vec{r}_{OB} \times \vec{F}_2 \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.15 & 0.1 & 1 \\ -300 \sin \theta & 300 \cos \theta & 0 \end{vmatrix}\end{aligned}$$

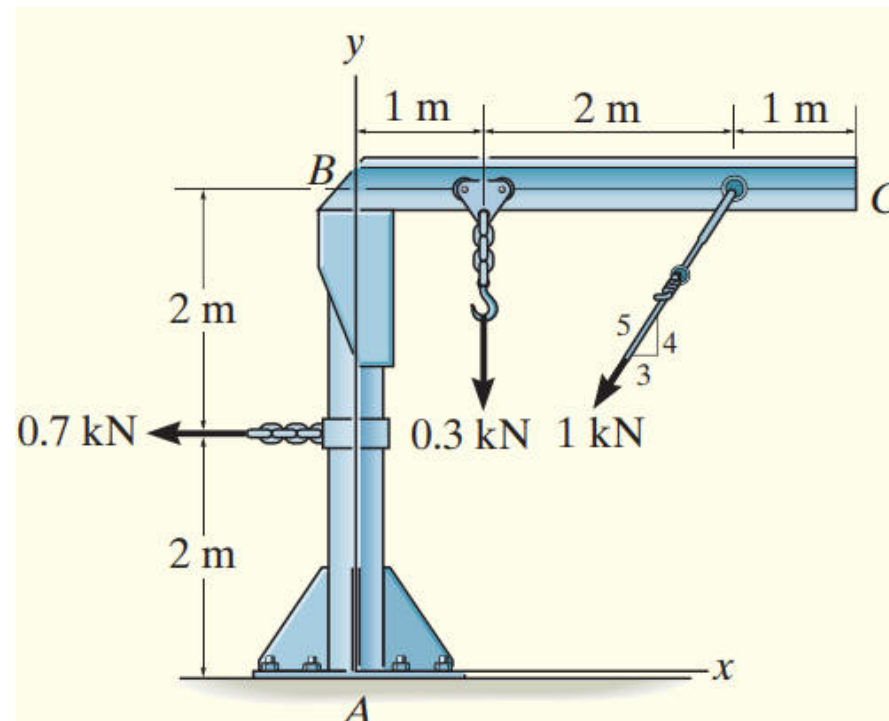


$$\begin{aligned}M_O &= \vec{M} + \vec{M}_{F_2} \\ \vec{F}_R &= \vec{F}_1 + \vec{F}_2\end{aligned}$$

A diagram shows a point O with two green arrows originating from it, representing the resultant force and couple moment.

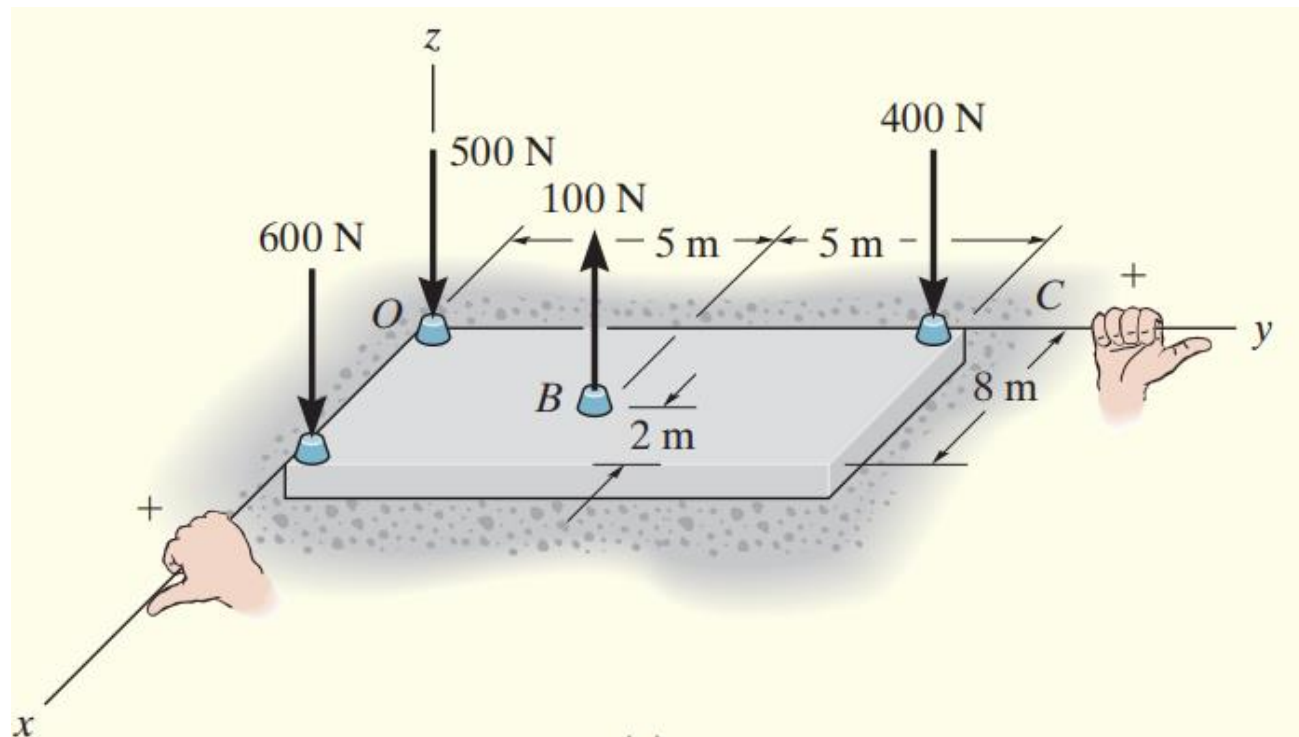
Practice at Home

5. The jib crane shown below is subjected to three coplanar forces. Replace this loading by an equivalent resultant force and specify where the resultant's line of action intersects the column AB and boom BC .



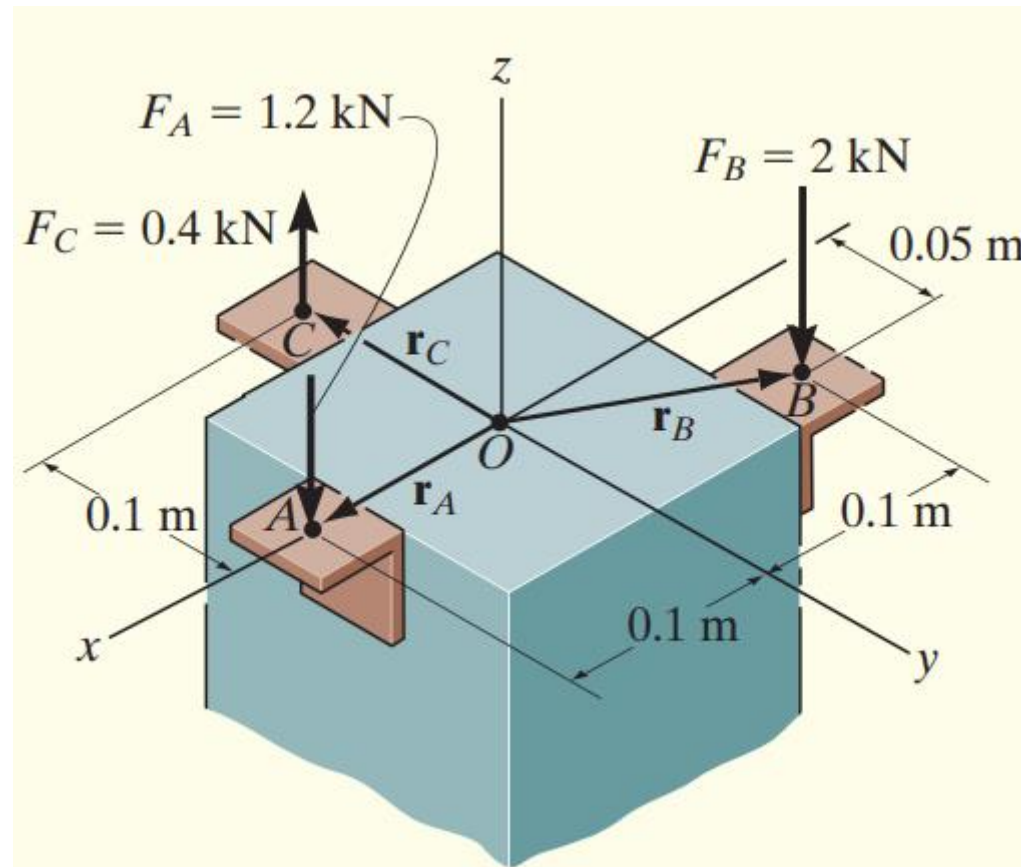
Practice at Home

6. The slab shown below is subjected to four parallel forces. Determine the magnitude and direction of a resultant force equivalent to the given force system and locate its point of application on the slab.



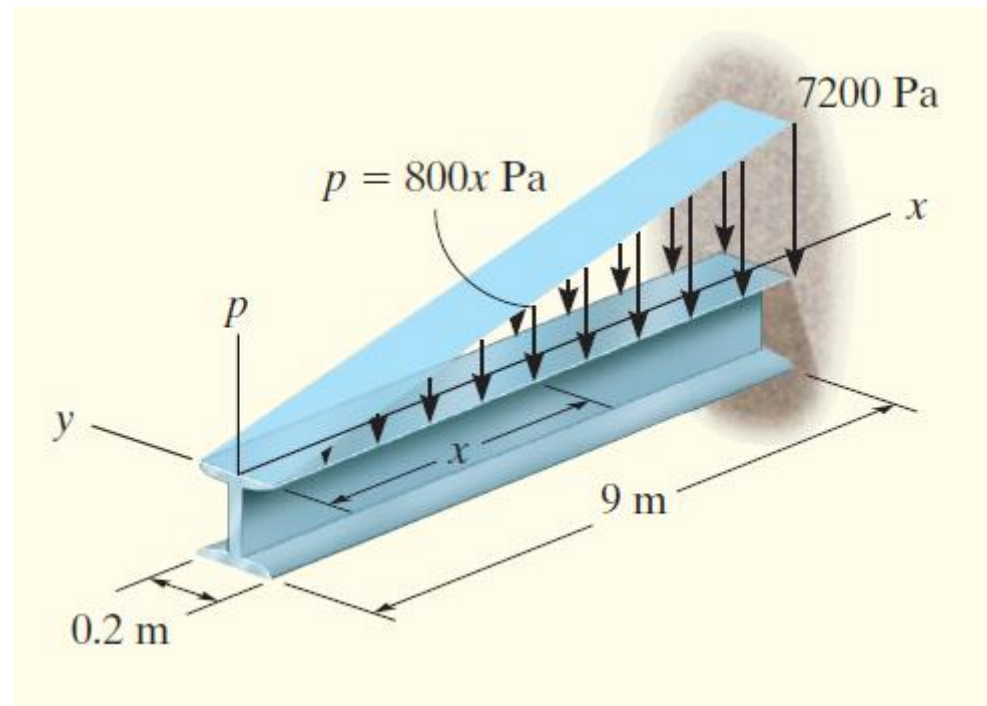
Practice at Home

7. Replace the force system shown below by an equivalent resultant force and specify its point of application on the pedestal.



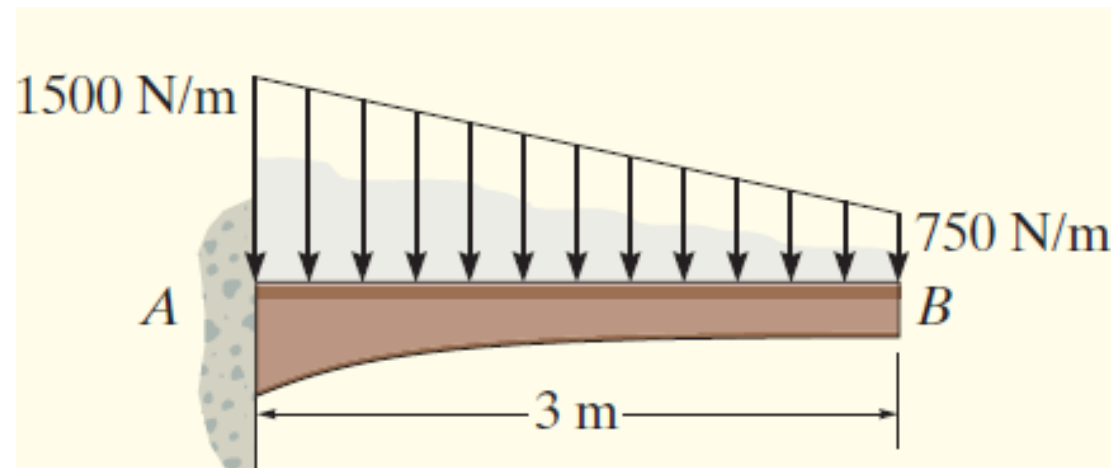
Practice at Home

8. A distributed loading of $p = 180x$ Pa acts over the top surface of the beam shown below. Determine the magnitude and location of the equivalent resultant force.



Practice at Home

9. The granular material exerts the distributed loading on the beam as shown below. Determine the magnitude and location of the equivalent resultant of this load.



Practice at Home

10. Arm AB rotates in the range and the spring is unstretched when $\theta = 90^\circ$. Determine as a function of θ , the moment about O of the spring force as applied at B . Plot the three scalar components of and state the maximum absolute value of each component.

$$\vec{M}_{F_O} = \vec{r} \times \vec{F}$$

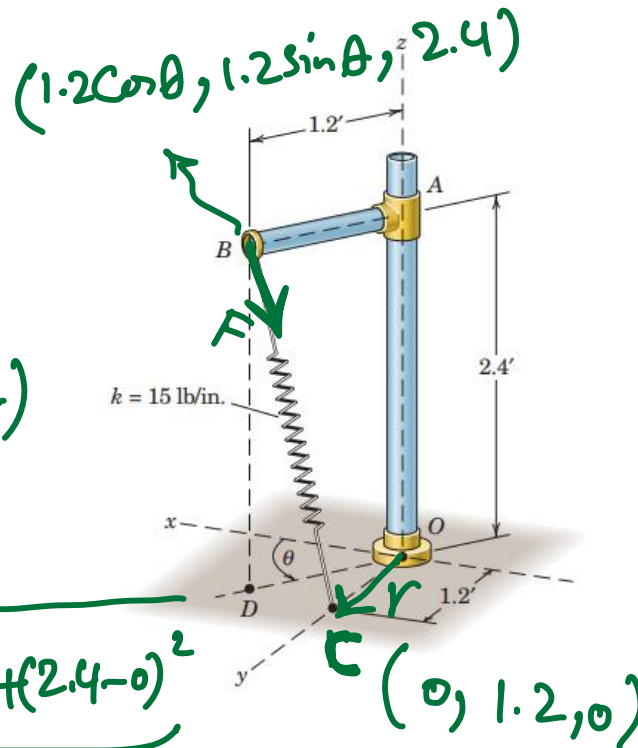
$$\vec{r} = 1.2 \hat{j} \text{ [in]}$$

$$F = Ks = Kf(\theta) \text{ (scalar)}$$

$$s = |\vec{r}_{B/C}| - l_0$$

$$|\vec{r}_{B/C}| = \sqrt{(1.2 \cos \theta - 0)^2 + (1.2 \sin \theta - 1.2)^2 + (2.4 - 0)^2}$$

$f(\theta)$



$$\checkmark \theta = 90^\circ$$

$$B(0, 1.2, 2.4)$$

$$2.4' = l_0$$

$$\vec{F} = Kf(\theta) \hat{u}_{r_{B/C}}(\theta)$$